

Shock Theory in a Supersonic Wind Tunnel

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1 Introduction

This experiment demonstrates the shock and expansion structures that are formed around a wedge airfoil in a supersonic tunnel. We used a setup involving a system that accelerates air to supersonic speeds and forces it across a wedge airfoil, thereby creating a shock and an expansion fan of notable characteristics that we may compare to the predictions of the theory. This system was set up in a lab at Monash University with access supplied to students for this study. The Schlieren imagery technique¹ (Yamamoto 2012) was used to image the shocks formed around the airfoil.

We will be analysing the oblique shock and expansion fan that is formed, using both the normal shock relations with the normalised mach numbers, as well as the Prandtl-Meyer expansion fan equations. We will perform a digital image analysis to achieve this².

We can construct a set of predictions for the mach numbers at three different locations: upstream (Region 1), between the shock and expansion fan (Region 2), and after the expansion fan (Region 3). The first mach number can be predicted using the compressible area ratio formula, where A^* is the neck area ($A^*=2560\text{mm}^2$) and A is the area at the wedge location ($A=6400\text{mm}^2$). We must assume air is an ideal and calorically perfect gas, where $\gamma = 1.4$.

$$\left(\frac{A}{A^*}\right)^2 = \frac{1}{M_1^2} \left(\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_1^2 \right) \right)^{\frac{\gamma+1}{\gamma-1}} \quad (1)$$

The second mach number is found using normal shock relations. We need to normalise the mach number such that we use the magnitude normal to the shock angle. We must acquire the predicted beta values of the wedge (the angle between the shock and the horizontal)⁴. This is accomplished using the Theta-Beta-Mach equation. We did this for six different wedge angles (-10, -5, 0, 5, 10, 15 degrees).

$$\tan(\theta) = 2 \cot(\beta) \frac{M_1^2 \sin^2(\beta) - 1}{M_1^2 (\gamma + \cos(2\beta)) + 2} \quad (2)$$

With a β value we may normalise our mach numbers around the shock formed. Denoting normal (n) mach numbers as $M_{i,n}$,

$$M_{2,n}^2 = \frac{1 + \left(\frac{\gamma-1}{2}\right) M_{1,n}^2}{\gamma M_{1,n}^2 - \frac{\gamma-1}{2}} \quad (3)$$

The third mach number occurs over an expansion fan. Assuming an ideal Prandtl-Meyer expansion fan, we can make a prediction based on the equation

$$\theta = \nu(M_3) - \nu(M_2) \quad (4)$$

The full $\nu(M)$ equation is in the appendix (Equation 7). Once we produced a set of predicted mach numbers for each of our three locations, we performed the wind tunnel experiment and measured the angles to find the experimental mach numbers. Our results yielded some correlation, but interestingly the third mach number was the most unpredictable result that occurred in the experiment.

2 Results and Discussion

The Schlieren imaging of the airfoil wedge produced over 1,500 images over a few seconds of footage. We found some noticeable randomness and fluctuations that occurred due to the rapid movement of air over the wedge. To accurately measure the angles from the images in ImageJ, we needed to construct an 'average' image for each of the six angles. This was done efficiently using a Python script that segmented the over 1,500 images into six different index ranges (excluding the transition images where the wedge was moving between angles), and then taking average pixel values across the entire range to construct a completely new 'mean' image that represented the target angle. Figure 1 demonstrates how this affects our results.

¹An optical technique used to capture shocks by visualising the changes in refractive index (density) of a fluid, performed using a collimated light beam and sensitive mirror-detection.

²We used the ImageJ application to perform this, allowing us to accurately determine shock angles.

³This results in $M_1=2.44$

⁴See appendix for table of angles

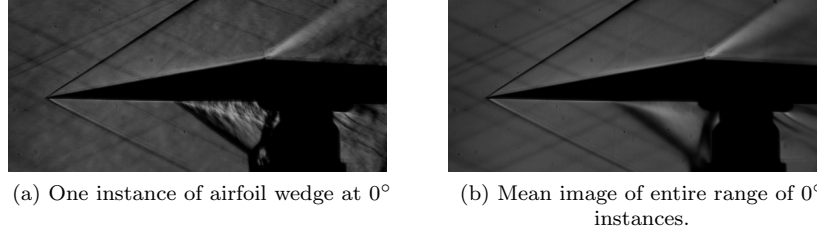


Figure 1: Comparison between one instance of the wedge and the mean image. We can see a much smoother end result, minimising the effects of air viscosity and producing much clearer shock and expansion fan lines.

Our predicted M_1 value was constant at 2.44. This makes sense as the speed of the incoming air does not depend on its flow over the wedge airfoil. When we performed the experiment, we found that the M_1 value was consistently lower than the predicted value. This is seen in Figure 2.

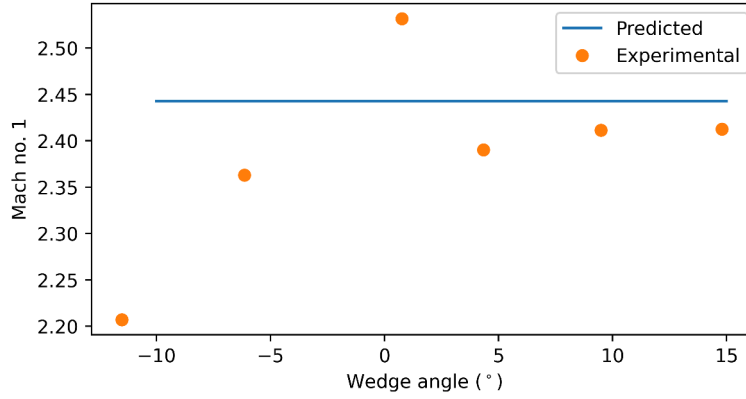


Figure 2: Upstream Mach number plotted against wedge angle. Pictured are the experimental results for six different initial wedge angles, as well as the theoretical mach number.

In the appendix there is a set of sample calculations for how we used the measured angles from ImageJ and converted this into an experimental mach number value. As we can see in the figure, the mach number does not significantly vary with respect to wedge angle. This is an expected result, as predicted by the theory. The most notable feature of the plot is that our experimental mach numbers are consistently below the theoretical mach number. When we made our prediction, we made a series of notable assumptions. We assumed inviscid flow in only two dimensions across the airfoil, and most importantly, uniform flow. A notable point of contention that suggests this assumption is not upheld in reality can be found by looking closely at the mean image in Figure 1. There are several more less prominent lines in the image. These occur as a result of shock reflections off the boundary walls of the supersonic tunnel. Over a shock, the mach number will decrease. Since there are several much weaker shocks between where the air enters the tunnel and the point it reaches the airfoil, the air might have decreased in speed to quite a high statistical significance. This is the most likely cause of the lower mach numbers that we found.

The second mach number was predicted using the normal shock relations. We needed to be careful and ensure we normalised our mach numbers. To achieve this, we needed to determine a set of predicted β values. Given M_1 and θ , this was simply accomplished using Equation 2. With these β values, we can use simple trigonometry to normalise M_1 ⁵, then apply the normal shock relation in Equation 3, and lastly normalise the result⁶ to find M_2 . Our experimental M_2 values were determined using expansion fan theory (Refer to Figure 7). We can use our measured angles in Equation 5 to determine M_2 . Both the predicted and experimental mach numbers are plotted in Figure 3.

⁵ $M_{1,n} = M_1 \sin(\beta)$

⁶ $M_2 = \frac{M_{2,n}}{\sin(\beta - \theta)}$

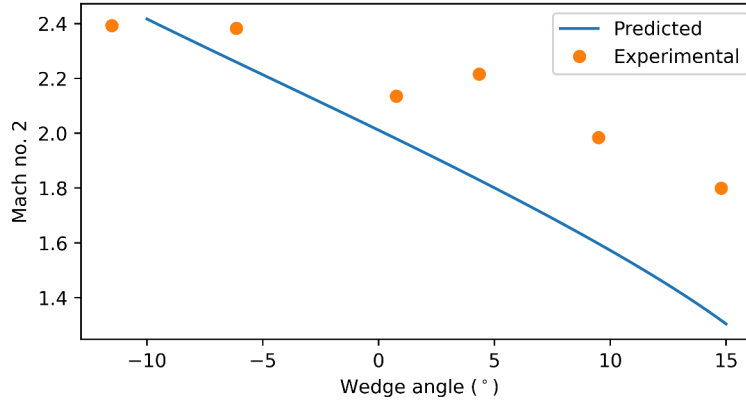


Figure 3: Second Mach number (speed between wedge and corner) plotted against wedge angle. Pictured are the experimental results for six different initial wedge angles, as well as the theoretical mach number.

The experimental data follows the downward trend that we see from the prediction, however it does not fall off nearly as rapidly. While there is an initial correlation, the experimental values do not decrease as rapidly and are, as a result, much higher than the prediction at higher wedge angles. A significant point of error in this data arises from our measurement of the expansion fan angles. Unlike a shock, which is a near-instantaneous change in the properties of the air, an expansion fan is a continuous change in properties. This makes determining an exact line along which the air properties change to be highly ambiguous. In addition to this, the Prandtl-Meyer expansion relation, dependent on μ_1 , is highly sensitive to small changes in angle. A small discrepancy in the measurement will result in a very different mach number. Another possible explanation is due to the shock being weaker than anticipated. This isn't as likely given the strong β correlation in Figure 5, however due to the presence of the shock reflections discussed earlier, it may be possible that the shock is not as strong, resulting in a smaller decrease in mach number after the shock.

We also measured μ_2 between the wedge surface and the trailing edge of the expansion fan. This angle allowed us to determine the experimental third mach number using the Prandtl-Meyer relations. The prediction for M_3 is found using the expansion fan equation (Equation 4), where M_2 is a given variable and $\nu(M)$ is given in full form in Equation 7. We needed to use a more involved root-finding algorithm to solve this equation due to its complexity. The results of both the prediction and the experiment can be compared in Figure 4.

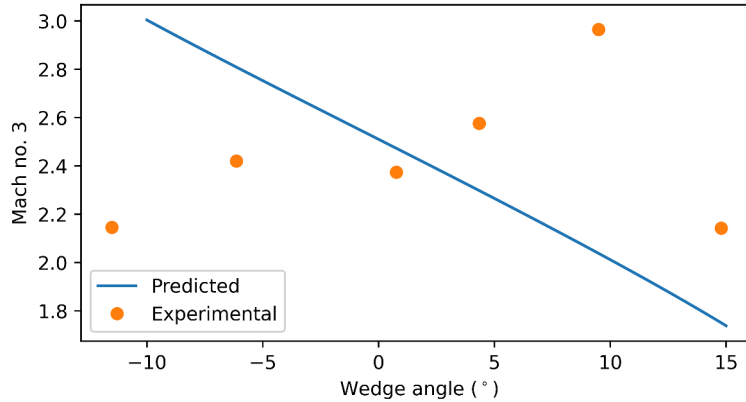


Figure 4: Third Mach number (post-expansion fan) plotted against wedge angle. Pictured are the experimental results for six different initial wedge angles, as well as the theoretical mach number.

There are two important features present in this figure. The first, encouragingly, is that our values for M_3 are generally higher than they are for M_2 . This aligns with the main feature of an expansion fan, which is the increase in mach number. Conversely, an oblique shock will lower the mach number, which is generally the case in our data across M_1 and M_2 . However, most notably, we see almost no correlation between the experimental and predicted results for M_3 . The measurement for μ_2 was highly inaccurate, even more so than for μ_1 . The

trailing edge of the expansion fan is extremely faded and difficult to determine, meaning we may end up with extremely large deviations from the prediction. Despite there being no clear correlation between our results and the theory, this does not suggest anything is incorrect with the predicted results. Due to the high degree of uncertainty and inaccuracy in our data, it is not appropriate to determine a conclusion from the results for M_3 .

We can, however, say that the oblique shock theory matched what we observed in our experiment. If we take a look at the predicted and experimental β angles, we can see a very close correlation. See Figure 5.

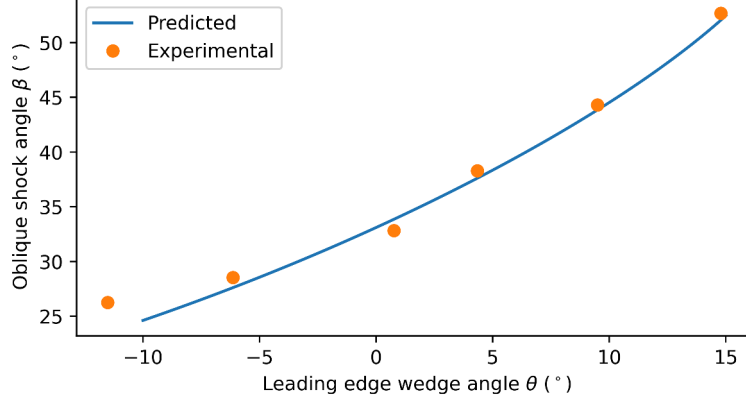


Figure 5: Oblique shock angle β against leading edge angle θ of airfoil wedge. Both predicted and experimental results are included.

The predicted and experimental β values are extremely close, suggesting our predictions for the β values surrounding the oblique shock were highly accurate.

The results exhibited a high degree of uncertainty that was mostly due to the difficulty and ambiguity in measuring μ angles around the expansion fan. As a result, we saw a higher degree of correlation between predicted and experimental results over the oblique shock, but very little correlation with the predictions made using the Prandtl-Meyer expansion relations. A similar study was done by Khan (2019) whereby mach number, as well as characteristics including pressure, temperature, and density were measured over an oblique shock. Similar discrepancies were found and the reasons were discovered to be due to the inviscid assumption as well as the boundary layer interaction with the flow (uniform flow assumption does not hold).

In addition to the uncertainties mentioned including the shock reflections, measurement difficulties, and non-ideal assumptions (inviscid, uniform flow), it is also important to take into account the three-dimensional nature of the wedge. Our results were predicted using two-dimensional approximations. A shock around a cone is weaker than a shock over an ideal wedge (Heli-air.net 2016), and since our wedge has depth in the third dimension, it follows that the oblique shock is likely to be weaker in reality than what we have predicted.

3 Application

This indirect method of measuring the mach numbers using the shock and expansion fan angles is not as ideal as it could be. A better way to calculate the velocity of air in each of the three regions would be to make use of Particle Image Velocimetry (Harte 2024). This is an optical method of precise particle tracking using lasers. It involves using a CCD camera with a high powered laser pointing through the transparent supersonic tunnel walls. There must be a small injection device installed into the tunnel that releases light-scattering particles. The material must not significantly affect the properties of the air, and the most common microspheres⁷ used include fluorescent polymer microspheres with a $6\mu\text{m}$ diameter. More information on equipment can be found in the Appendix.

The method works by determining the distance between groups of particles (a few square pixels) between consecutive frames, given some time difference between each image. This process is known to be computationally intensive. The governing equations of this method can be complex depending on the application. Typically, the system creates a displacement field and divides it by the time between frames to determine a velocity field.

⁷A term used to refer to miniscule spherical-shaped particles

Then, the x and y components are separated. PIV can also be applied to three dimensions, but this is more complicated than we need.

The velocity is then:

$$v = \sqrt{v_x^2 + v_y^2}, v_x = \frac{\Delta x}{\Delta t}$$

Then, given the speed of sound in air (as an ideal gas where $\lambda = 1.4$, $R = 287 \text{Jkg}^{-1}$):

$$a = \sqrt{\gamma RT},$$

we may determine mach number

$$M = \frac{v}{a}.$$

This method is preferable as it is non-intrusive. There is no need for large probes to be fitted inside the tunnel that require large changes to the apparatus. Most of the analysis can be done in real time with a computer as well, meaning large amounts of data can be extracted in a short period of time without much trouble after the system has been set up. The method is also highly accurate.

The most notable drawback of this method is that there is a nonzero possibility that the microspheres may influence the properties of the airflow. Another drawback is that the CCD must be sufficiently far from the tunnel such that the effect of parallax is negligible, meaning the system might need to be housed over quite a large area. Overall, this method is definitely an efficient way to directly determine the velocities and mach numbers of the air inside the tunnel.

4 Conclusion

The oblique shock theory closely predicts the behaviour that we saw in our experiment when determining the mach numbers in each of the regions. The theoretical value of M_1 at 2.44 was quite closely matched by the experimental values, albeit slightly slower, and the downward trend predicted for M_2 was observed in reality, however the drop off was not as extreme as predicted. The third mach number did not yield much correlation due to the ambiguity of measuring the expansion fan angles.

We found that the expansion fan theory made use of assumptions that did not entirely hold throughout our experiment, which led to some discrepancies between our experimental and predicted results for the final mach number. Most notably the inviscid flow assumption and the uniform flow assumption. We encountered the presence of shock reflections which weakened the upstream mach number, as well as the viscous nature of the air weakening our shock and expansion fan which affected our results. In addition to this, the human error introduced with measuring angles around the expansion fan caused much higher errors resulting from the difficulty in finding the exact edges of the expansion fan.

One major way in which this experiment could be improved would be to take 3-5 measurements of the experiment, and taking more than 1,500 frames of the air movement. Doing this would allow us to observe standard deviation error between measurements, and include plots with error bars to create much more conclusive figures that more accurately describe what occurred in the experiment. Performing the experiment once made us susceptible to significant outliers that may have impacted results. Overall, this was a highly interesting study of the oblique shock and expansion fan theory and how it applies to an experimental scenario.

5 References

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- Harte, N.C., Obrist, D., Versluis, M., E. Groot Jebbink, M. Caversaccio, Wimmer, W. and G. Lajoinie (2024). Second order and transverse flow visualization through three-dimensional particle image velocimetry in millimetric ducts. Experimental Thermal and Fluid Science, 159, pp.111296–111296. doi:weblink.

6 Appendix

6.1 Appendix 1: Sample Calculations

Using Equation 1, we determine that the predicted value of M_1 is 2.44. In our experimental analysis, we determined M_1 solely by θ and the resultant β angles, using Equation 2, the Theta-Beta-Mach equation. Our experimental angle values are listed in Table 1. Taking the example of $\theta = 0.76^\circ$ and $\beta = 32.83^\circ$:

$$\tan(0.76^\circ) = 2 \cot(32.83^\circ) \frac{M_1^2 \sin^2(32.83^\circ) - 1}{M_1^2(\gamma + \cos(2 \cdot 32.83^\circ)) + 2}$$

Given $\gamma = 1.4$, we used numpy `fsolve` to numerically solve the function for M_1 .

```
import numpy as np
exp_angles = [0.76, -11.52, -6.14, 4.34, 9.49, 14.79]
exp_theta_1 = exp_angles + 10.62
exp_beta_1 = [32.83, 26.27, 28.53, 38.30, 44.31, 52.69]
gamma = 1.4

for i in range(len(exp_angles)):
    def f_betamach_exp(M):
        return (2/np.tan(np.radians(exp_beta_1[i]))*(M**2*(np.sin(np.radians(
            exp_beta_1[i]))**2)-1)/(M**2*(gamma+np.cos(2*np.radians(exp_beta_1[i]
            )))+2)-np.tan(np.radians(exp_theta_1[i]))

    exp_M_1[i] = fsolve(f_betamach_exp, 2.4)[0]
```

In this example, M_1 comes out to be 2.53. To find M_2 and M_3 , the prediction follows the normal shock relations and the Prandtl-Meyer expansion fan equation. To achieve this experimentally, we already have the angles μ required to do this independently from the images alone. From the theory, it follows that

$$M_2 = \frac{1}{\sin(\mu_1)} \quad (5)$$

$$M_2 = \frac{1}{\sin(27.92^\circ)}$$
$$M_2 = 2.14.$$

And further:

$$M_3 = \frac{1}{\sin(\mu_2)} \quad (6)$$

$$M_3 = \frac{1}{\sin(24.91^\circ)}$$
$$M_3 = 2.37.$$

6.2 Appendix 2: Prandtl-Meyer Expansion Fan

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \arctan\left(\sqrt{\frac{\gamma-1}{\gamma+1}}(M^2-1)\right) - \arctan(\sqrt{M^2-1}) \quad (7)$$

6.3 Appendix 3: Diagrams

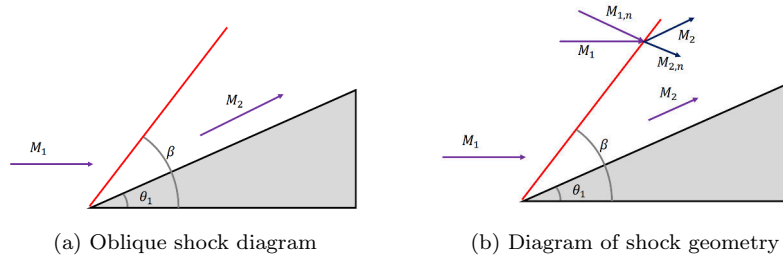


Figure 6: Diagrams of oblique shock geometry

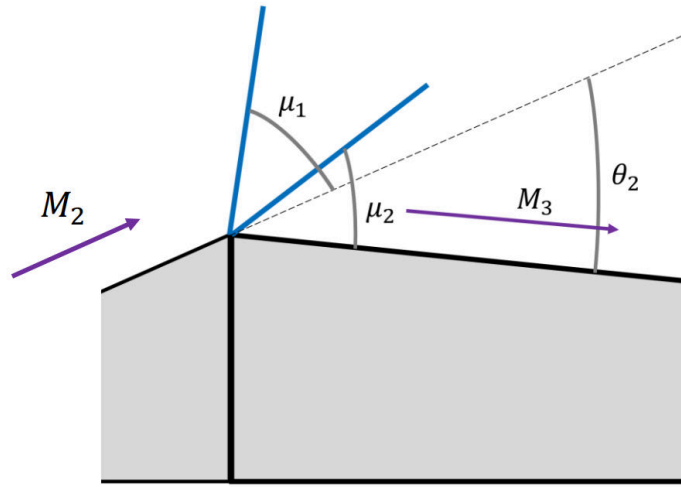


Figure 7: Prandtl-Meyer expansion fan geometry

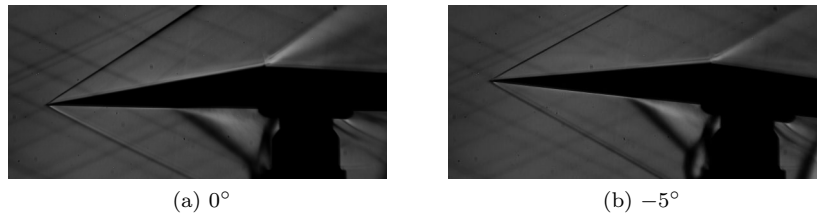


Figure 8: Experimental mean images of airfoil wedge taken using the Schlieren imagery technique.

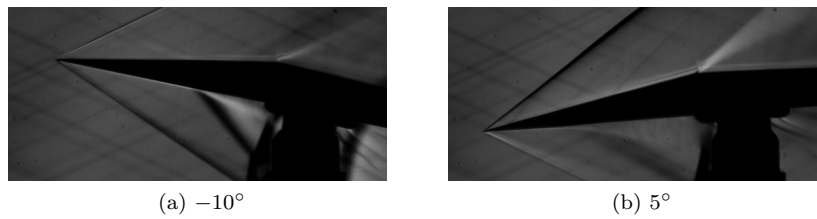


Figure 9: Continued.



Figure 10: Continued.

6.4 Appendix 4: Table of values

Wedge angle α (°)	β (°)	μ_1 (°)	μ_2 (°)	M_1	M_2	M_3
-11.52	26.27	24.71	27.78	2.21	2.39	2.15
-6.14	28.53	24.81	24.40	2.36	2.38	2.42
0.76	32.83	27.92	24.91	2.53	2.14	2.37
4.34	38.30	26.83	22.84	2.39	2.22	2.58
9.49	44.31	30.26	19.71	2.41	1.98	2.97
14.79	52.69	33.77	27.82	2.41	1.80	2.14

Table 1: Table of measured angles and mach numbers, determined experimentally using ImageJ and calculated using the Theta-Beta-Mach equation (Equation 2).

6.5 Appendix 5: Application

The most common laser wavelengths used for particle image velocimetry are any harmonic of 1064nm (most commonly 532nm, or green visible light) produced by a neodymium-doped yttrium aluminium garnet crystal (Nd-YAG laser) (Harte 2024). A CCD (charge coupled device) needs to be used with an extremely high resolution. These are used as they have high refresh rates and can capture large amounts of frames in a short space of time.

6.6 Appendix 6: Python code analysis

```
import numpy as np
from scipy.optimize import fsolve
import matplotlib.pyplot as plt

## Code to make predictions based on ideal theory and compares to experiment
# Written by: Daniel Smith

## Problem Setup
# Here we set up the problem with known variables

# Define wedge angles to predict
wedge_angles = np.radians(np.linspace(-10, 15, 100))

# Givens, fill these with known values
A_star = 0.00256
A = 0.0064
wedge_theta_1 = np.radians(10.62)
wedge_theta_2 = np.radians(12.67)

gamma = 1.4

## Calculations
# This section performs the calculations using ideal theory

# Initialise vectors
theta_1 = np.zeros(len(wedge_angles))
theta_2 = np.zeros(len(wedge_angles))
M_1 = np.zeros(len(wedge_angles))
M_2 = np.zeros(len(wedge_angles))
M_3 = np.zeros(len(wedge_angles))
beta_1 = np.zeros(len(wedge_angles))
mu_1 = np.zeros(len(wedge_angles))
mu_2 = np.zeros(len(wedge_angles))

# Loop through all the wedge angles and calculate all variables for each angle
for i in range(len(wedge_angles)):
    # How does theta_1 and theta_2 change with wedge angle?
    theta_1[i] = wedge_angles[i] + wedge_theta_1 # Angle of leading edge
    theta_2[i] = wedge_theta_2 # Angle of top of wedge

    # === Tunnel Mach Number ===
    # Define a function  $f(x) = 0$  that we will solve for  $x$ , in our case 'f' is
    the area ratio equation and 'x' is the mach number
    def f_arearatio(M):
        return ((gamma + 1) / 2) ** (-(gamma + 1) / (2 * (gamma - 1))) * 1 / M *
            (1 + (gamma - 1) / 2 * M ** 2) ** ((gamma + 1) / (2 * (gamma - 1)))
            - A / A_star
    # Numerically solve the equation using fsolve with an initial guess of M=2.4
    M_1[i] = fsolve(f_arearatio, 2.4)[0]

    # === Leading Shock Angle ===
    # Define a function  $f(x) = 0$ 
    def f_betamach(beta):
```

```

        return (2/np.tan(beta))*(M_1[i]**2*(np.sin(beta)**2)-1)/(M_1[i]**2*(
            gamma+np.cos(2*beta))+2)-np.tan(theta_1[i])
# Numerically solve the equation using fsolve with an initial guess
beta_1[i] = fsolve(f_betamach, 2*wedge_theta_1)[0]

# === Post-Shock Mach Number ===
# Use oblique and normal shock relations to find the mach number post-shock
M_n1 = M_1[i]*np.sin(beta_1[i])
M_n2 = np.sqrt((1+((gamma-1)/2)*M_n1**2)/(gamma*M_n1**2-(gamma-1)/(2)))
M_2[i] = M_n2/(np.sin(beta_1[i]-theta_1[i]))

# === Leading expansion fan angle ===
# Use expansion wave theory to find the leading edge angle of the expansion
# fan
mu_1[i] = np.arcsin(1/M_2[i])

# === Post-fan Mach number ===
# Use Prandtl-meyer theory to find nu_2
nu_2 = np.sqrt(((gamma+1)/(gamma-1))*np.arctan(np.sqrt(((gamma-1)/(gamma+1))
    *(M_2[i]**2-1)))-np.arctan(np.sqrt(M_2[i]**2-1)))
# Combine nu_2 with theta to find nu_3
nu_3 = theta_2[i] + nu_2

# Define a function f(x) = 0
def f_prantlmeyer(M):
    return np.sqrt(((gamma+1)/(gamma-1))*np.arctan(np.sqrt(((gamma-1)/(gamma
        +1))*(M**2-1)))-np.arctan(np.sqrt(M**2-1))) - nu_3
# Numerically solve the equation using fsolve with an initial guess
M_3[i] = fsolve(f_prantlmeyer, 3)[0]

# === Trailing expansion fan angle ===
# Use expansion fan theory to find the angle of the trailing edge of the
# expansion fan
mu_2[i] = np.arcsin(1/M_3[i])

# Experimental Data
# Input values from experimental analysis
exp_angles = [0.76, -11.52, -6.14, 4.34, 9.49, 14.79]
exp_beta_1 = [32.83, 26.27, 28.53, 38.30, 44.31, 52.69]
exp_mu_1 = [27.92, 24.71, 24.81, 26.83, 30.26, 33.77]
exp_mu_2 = [24.91, 27.78, 24.40, 22.84, 19.71, 27.82]

# Initialise vectors
exp_theta_1 = np.zeros(len(exp_angles))
exp_theta_2 = np.zeros(len(exp_angles))
exp_M_1 = np.zeros(len(exp_angles))
exp_M_2 = np.zeros(len(exp_angles))
exp_M_3 = np.zeros(len(exp_angles))

for i in range(len(exp_angles)):
    # How does theta_1 and theta_2 change with wedge angle?
    exp_theta_1[i] = exp_angles[i] + np.degrees(wedge_theta_1) # Angle of
        leading edge
    exp_theta_2[i] = wedge_theta_2 # Angle of top of wedge

```

```

# === Pre-Shock Mach Number ===
# Define a function  $f(x) = 0$  that we will solve for  $x$ 
def f_betamach_exp(M):
    return (2/np.tan(np.radians(exp_beta_1[i])))*(M**2*(np.sin(np.radians(
        exp_beta_1[i]))**2)-1)/(M**2*(gamma+np.cos(2*np.radians(exp_beta_1[i]
        )))+2)-np.tan(np.radians(exp_theta_1[i])))
# Numerically solve the equation using fsolve with an initial guess
exp_M_1[i] = fsolve(f_betamach_exp, 2.4)[0]

# === Pre-fan Mach number ===
# Use expansion fan theory to find the mach number
exp_M_2[i] = 1 / np.sin(np.radians(exp_mu_1[i]))

# === Post-fan Mach number ===
# Use expansion fan theory to find the mach number
exp_M_3[i] = 1 / np.sin(np.radians(exp_mu_2[i]))

## Plotting

# Generate the required plots

# Predicted plots

plt.figure(1).set_figheight(3.2)
plt.plot(np.degrees(wedge_angles), M_1, label="Predicted")
plt.plot(exp_angles, exp_M_1, linestyle="none", marker="o", label="Experimental")
plt.xlabel(r"Wedge-angle-($^\circ$)")
plt.ylabel(r"Mach-no.-1")
plt.legend()
plt.savefig("m1.png", bbox_inches='tight', dpi=500, format='png')

plt.figure(2).set_figheight(3.2)
plt.plot(np.degrees(wedge_angles), M_2, label="Predicted")
plt.plot(exp_angles, exp_M_2, linestyle="none", marker="o", label="Experimental")
plt.xlabel(r"Wedge-angle-($^\circ$)")
plt.ylabel(r"Mach-no.-2")
plt.legend()
plt.savefig("m2.png", bbox_inches='tight', dpi=500, format='png')

plt.figure(3).set_figheight(3.2)
plt.plot(np.degrees(wedge_angles), M_3, label="Predicted")
plt.plot(exp_angles, exp_M_3, linestyle="none", marker="o", label="Experimental")
plt.xlabel(r"Wedge-angle-($^\circ$)")
plt.ylabel(r"Mach-no.-3")
plt.legend()
plt.savefig("m3.png", bbox_inches='tight', dpi=500, format='png')

plt.figure(4).set_figheight(3.2)
plt.plot(np.degrees(wedge_angles), np.degrees(beta_1), label="Predicted")
plt.plot(exp_angles, exp_beta_1, linestyle="none", marker="o", label="Experimental")
plt.xlabel(r"Leading-edge-wedge-angle-$\theta$-($^\circ$)")
plt.ylabel(r"Oblique-shock-angle-$\beta$-($^\circ$)")
plt.legend()
plt.savefig("beta.png", bbox_inches='tight', dpi=500, format='png')

```